Impact of intensified dietary therapy on energy and nutrient intakes and fatty acid composition of serum lipids in patients with recently diagnosed non–insulin-dependent diabetes mellitus

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ABSTRACT: Compliance with dietary recommendations and the effect of intensified dietary therapy on energy and nutrient intakes and fatty acid composition of serum lipids were studied in 86 obese subjects (aged 40 to 64 years) with recently diagnosed non–insulin-dependent diabetes mellitus (NIDDM). After three months of basic education, the subjects were randomly separated into an intervention group (n = 40) and a conventional treatment group (n = 46). Members of the intervention group participated in 12 months of intensified education; those in the conventional group visited local health centers. Compliance with dietary instructions was monitored through food records. Intensified dietary therapy resulted in greater weight loss, better metabolic control, and a less atherogenic lipid profile than conventional treatment. Intake of energy and saturated fatty acids tended to decline in the intervention group. A higher percentage of patients in the intervention group had a total fat intake of 30% of energy or less after 15 months (32.5% [12 of 38] vs 17.4% [8 of 46]). Similarly, more patients in the intervention group had a saturated fatty acid intake of 10% or less of total energy intake at the end of the study (35.0% [13 of 38] vs 8.7% [4 of 46]). The mean dietary cholesterol intake was within recommendations in both groups at the end of the study. The relative percentage of linoleic acid of serum lipids increased significantly and the relative percentage of palmitic acid of serum triglycerides, phospholipids, and cholesterol esters decreased in the intervention group. These changes indicate that intensified dietary therapy improved the quality of fat in the diet of patients with NIDDM. *J Am Diet Assoc. 1993; 93:276-283.*

The main goals of dietary therapy for obese patients with non–insulin-dependent diabetes mellitus (NIDDM) are weight reduction, correction of hyperglycemia, and a decrease in atherosclerotic risk factors, that is, correction of dyslipidemias and elevated blood pressure (1-5). Most studies of the diets of patients with NIDDM have concentrated on weight reduction and correction of hyperglycemia (6-8). Some studies have, however, examined the composition of diets and the effect of dietary education on nutrient and fatty acid composition of the diet. In these studies, the mean fat intake varied between 30% and 45% of total energy (9-13), and frequently the diets were rich in saturated fatty acids. Carbohydrate content and fiber content were lower than recommended (10,11,13), but the sucrose content was within recommendations (11).

On the basis of these studies, it seems that current dietary recommendations cannot be easily applied in the clinical setting. Fatty acid composition of serum lipids reflects dietary intake of fats in healthy subjects (14-17). However, there are only a few studies on fatty acid composition of serum lipids in patients with NIDDM, and even less is known about the changes in the composition during dietary therapy (18).

The aims of our study were (a) to examine the effects of intensified dietary therapy on food consumption patterns, energy and nutrient intakes, and the fatty acid composition of serum triglycerides, phospholipids, and cholesterol esters during dietary therapy in patients with recently diagnosed NIDDM; and (b) to determine how well the main goals of dietary recommendations are met by patients with recently diagnosed NIDDM.

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Table 1
Patient characteristics at the time non–insulin-dependent diabetes mellitus was diagnosed, history of cardiovascular diseases, and use of cardiovascular drugs at beginning of study

	Intervention group		Conventional treatment group	
	Men (*n*=21)	*Women* (*n*=19)	*Men* (*n*=28)	*Women* (*n*=18)
Patient characteristics				
Age (yr)	50.7±7.7	53.7±6.3	54.0±6.6	54.4±6.4
Weight (kg)	98.1±13.0	88.2±15.3	97.7±12.7	87.7±17.8
Height (cm)	173.4±4.8	158.5±6.2	174.7±5.4	157.9±6.9
Body mass index[a]	32.6±3.9	35.3±6.7	32.0±3.4	35.1±6.0
Fasting blood glucose level (mmol/L)[b]	11.3±3.3	11.3±5.7	11.5±3.2	12.1±4.1
	total		*total*	
Cardiovascular diseases				
Myocardial infarction	3		3	
Angina pectoris	7		8	
Cardiac failure	4		4	
Hypertension	19		21	
Stroke	0		1	
Claudication	2		0	
Use of cardiovascular drugs[c]	25		19	

[a]Weight (kg)/height (m)^2.
[b]To convert mmol/L glucose to mg/dL, multiply mmol/L by 18.0. To convert mg/dL glucose to mmol/L, multiply by 0.0555. Glucose of 6.0 mmol/L = 108 mg/dL.
[c]Digitalis, β-blocker, diuretic, nitrates, calcium-channel blocker, or angiotensin-converting–enzyme inhibitor. Percent of total: intervention, 63%; conventional treatment, 41%.

SUBJECTS AND METHODS

Patient Characteristics and Study Procedure

Physicians working in five rural and one urban (Kuopio) health centers in the study area were asked to refer to the Kuopio University Hospital outpatient clinic all patients (40 to 64 years old) with newly diagnosed NIDDM and fasting blood glucose levels of 6.7 mmol/L or greater in repeated measurements.[1] All patients fulfilled the diagnostic criteria of the World Health Organization's Expert Committee for diabetes mellitus (19). Altogether, 108 patients were recruited, but 22 of them either did not fulfill the criteria for inclusion in the study or did not want to participate in the study. The final study population consisted of 86 obese patients. Patient characteristics at the time NIDDM was diagnosed are presented in Table 1. Because many patients had hypertension or coronary heart disease, cardiovascular medication was common. The study was approved by the Ethics Committee of Kuopio University Hospital. All patients gave oral consent before participating in the study.

A basic education was given to all participants in the study at 0 month (baseline), 6 weeks, and 3 months by a physician and a nurse specializing in diabetes education. These topics were discussed: the causes of NIDDM and its treatment modalities, follow-up measurements (eg, urinary glucose monitoring), and complications. The patients were advised to increase their physical activity; to restrict intakes of energy, fat—especially saturated fatty acids—and dietary cholesterol; to increase intakes of unsaturated fatty acids and unrefined carbohydrates; and to avoid large amounts of simple carbohydrates. The dietary instructions given by the nurse were tailored individually on the basis of knowledge from the patient's dietary history.

[1]To convert mmol/L glucose to mg/dL, multiply mmol/L by 18.0. To convert mg/dL glucose to mmol/L, multiply mg/dL by 0.0555. Glucose of 6.0 mmol/L = 108 mg/dL.

During the third visit (after 3 months) the patients were randomly separated into two treatment groups: intervention group (21 men and 19 women) and conventional treatment group (28 men and 18 women). After randomization, the patients in the conventional treatment group received the usual education given at the local health centers where they visited at 2- to 3-month intervals. In addition, patients in the conventional treatment group visited the outpatient clinic at 9 and 15 months from the baseline. Patients in the intervention group visited the outpatient clinic every second month—six times altogether. The intervention lasted 12 months. During each visit they met with a physician, a nurse specializing in diabetes, and a clinical nutritionist.

The following measurements were taken during each visit: weight, fasting blood glucose level, glycated hemoglobin (HbA_{1c}), and blood pressure. At 0, 3, 9, and 15 months, serum lipid levels were determined. Fatty acid composition of serum lipids was analyzed at baseline, 3 months, and 15 months.

Intensified Dietary Education

The intervention group received intensified dietary education from a clinical nutritionist. The goals of dietary therapy were weight reduction, normoglycemia, correction of dyslipidemias, and normalization of elevated blood pressure. The recommended diet was as follows: individually planned energy restriction; restriction of total fat (≤30% of energy), especially saturated fatty acids (≤10% of energy); restriction of dietary cholesterol (≤300 mg/day); a moderate increment of unsaturated fatty acids (≥20% of energy); and increased intake of foods containing unrefined carbohydrates, especially soluble fiber (eg, fruit, berries, and vegetables) (1). Intake of sucrose and other energy-containing sweeteners was not to exceed 10% of energy. Regular eating habits and moderate food consumption were emphasized.

To motivate the patients, the diet for each patient in the intervention group was tailored individually on the basis of

- Principles of the diabetic diet
- Fat in the diet—amount and quality
- Carbohydrates and fiber
- Use of sweeteners
- Advantages and disadvantages of special products for diabetic patients
- Behavior modification: how to motivate oneself; how to recognize situations in which it is difficult to control eating; what to do instead of eating; how to manage parties, traveling, and drinking alcohol
- Review of the most important things in the diet
- Food preparation in practice (group session)

FIG 1. Topics of intensified dietary education

food records. During each visit, the clinical nutritionist and the patient set one or two clear short-term goals for dietary change and a goal for weight loss. The educational topics are presented in Figure 1. To learn eating habits (eg, to notice situations in which control of eating disappears), the patients were asked to record eating habits (eg, where they ate, with whom they ate, and what else they did while eating).

Food Records

All patients completed 4-day (from Thursday through Sunday) food records using household measurements before 3-, 9-, and 15-month visits to the outpatient clinic. (Food records collected at 9 months were used only for diet counseling.) The patients had to fast overnight before coming to the outpatient clinic on Monday. Therefore, the records from Thursday to Saturday were included in the final analysis, which was done with the help of the Nutrica computer program (developed in 1989-1990 by the Social Insurance Institution of Finland) based on Finnish nutrient databases (20). Food records from two patients in the intervention group were completed improperly, so they were not included in the final analysis.

Anthropometric and Metabolic Measurements

Body weight was measured with an electrical scale. Patients fasted overnight and wore light-weight clothing. Standing height was measured (patients were not wearing shoes) and the readings were rounded up or down to the nearest whole number (1.0 cm).

Blood glucose level was determined by a glucose oxidase method (Glucose Auto/Stat HGA+1120 analyzer, Daiichi, Kyoto, Japan). Glycated HbA_{1c} was measured with commercial fast-liquid chromatography (Pharmacia Fine Chemical AB, Uppsala, Sweden).

Serum cholesterol and triglyceride levels were determined by enzymatic methods (21,22). High-density lipoprotein (HDL) cholesterol was determined after precipitation with dextran sulfate and magnesium chloride (23).

The fatty acid composition of serum lipids was determined by extracting the fat and separating cholesterol esters, phospholipids, and triglycerides with thin-layer chromatography. The transesterified fatty acids were analyzed by gas chromatography using a silica capillary column (24,25).

Statistical Methods

The statistical analyses were made using the SPSS-X program (1988, SPSS, Chicago, Ill)(26). All the variables were distributed normally. Multivariate analysis of variance for a repeated measures design was used to determine the changes between the measurements and the difference in the changes between the two groups in body weight, metabolic measurements, and fatty acid composition of serum lipids. If there were no statistically significant ($P>.05$) changes or differences, no further analysis was done. If there were significant changes over time, further analysis was done to determine whether the changes were significant for each group taken alone.

The changes in dietary variables between 3 and 15 months in each group were analyzed by paired t test. Analysis of variance was used to test differences between group means by gender in dietary variables at 3 or at 15 months.

The sample means of body weight, metabolic measurements, and the fatty acid composition of serum lipids were compared using the t test for independent samples at 15 months.

The proportions of patients who achieved the goals of metabolic measurements and diet at 3 and 15 months, as well as the changes in proportions between the measurements, were analyzed using the test of proportions based on standardized normal distribution (27).

The results for continuous variables are expressed as means ± standard deviation.

RESULTS

Anthropometric and Clinical Measurements

Body weight decreased significantly ($P<.01$) during the basic education period (0 to 3 months) in both groups. Patients in the intervention group continued to lose weight, whereas those in the conventional treatment group tended to gain weight during the intervention period (Table 2). The total mean weight reduction (CI = 95%) from the baseline (0 month) to the end of the study was −5.1 kg (−6.5 kg to −3.7 kg) in the intervention group and −2.2 kg (−3.8 kg to −0.6 kg) in the conventional treatment group ($P<.05$ between the groups).

The mean fasting blood glucose level decreased significantly ($P<.001$) in both groups between the time of diagnosis and 3 months. During the intervention period (3 to 15 months), the fasting blood glucose level decreased only in the intervention group. The fasting blood glucose level was significantly lower ($P<.05$) in the intervention group than in the conventional treatment group at the end of the study. Glycated HbA_{1c} decreased significantly ($P<.001$) in both groups during the basic education period and tended to decline during the intervention period. At the end of the study, glycated HbA_{1c} was lower ($P=.053$) in the intervention group than in the conventional treatment group.

Two patients in the intervention group and five patients in the conventional treatment group received sulfonylureas at the end of the study. Use of diuretics decreased and use of angiotensin-converting–enzyme inhibitors increased during the study. Nine patients in the intervention group with persistent hypercholesterolemia (serum total cholesterol ≥7 mmol/L at 9 months)[2] were treated with guar gum, and one patient received clofibrate because of hypertriglyceridemia (serum triglycerides ≥5 mmol/L at 6 months).[3] In the conventional treatment group, only one patient received guar gum. None of the drugs used could explain the differences in metabolic control between the groups.

[2]To convert mmol/L cholesterol to mg/dL, multiply mmol/L by 38.9. To convert mg/dL cholesterol to mmol/L, multiply by 0.026. Cholesterol of 5.00 mmol/L = 193 mg/dL.

[3]To convert mmol/L triglyceride to mg/dL, multiply mmol/L by 88.6. To convert mg/dL triglyceride to mmol/L, multiply mg/dL by 0.0113. Triglyceride of 1.80 mmol/L = 159 mg/dL.

Table 2
Mean values of body weight, glycemic control, and serum lipids at baseline, 3 months, and 15 months, and the change from 3 to 15 months

Variable/group[a]	0 mo	3 mo	15 mo	Change from 3 to 15 mo (mean; 95% CI)
	← mean ± standard deviation →			
Weight (kg)				
Intervention group	91.6±14.5	88.3±14.1^{**z}	86.5±13.7	−1.8 (−3.0-−0.5)
Conventional treatment group	92.2±14.7	88.8±14.0^{***z}	90.2±14.3	+1.0 (−0.1-+2.2)
Fasting blood glucose (mmol/L)[b]				
Intervention group	7.6±2.4	6.6±1.9^{***z}	6.2±1.8^{*y}	−0.4 (−0.9-+0.2)
Conventional treatment group	8.9±3.3	7.5±2.9^{***z}	7.5±2.2^{*x}	0.0 (−0.8-+0.8)
Glycated hemoglobin A_{1c} (%)				
Intervention group	8.4±2.2	7.1±1.8^{***z}	6.6±1.6	−0.6 (−1.2-−0.1)
Conventional treatment group	9.0±2.6	7.8±2.0^{***z}	7.5±1.7	−0.3 (−0.9-+0.2)
Serum cholesterol (mmol/L)[c]				
Intervention group	6.3±1.4	6.1±1.2	6.0±1.0	−0.1 (−0.3-+0.1)
Conventional treatment group	6.5±1.1	6.3±1.0	6.4±1.0	+0.1 (−0.2-+0.4)
Serum HDL-cholesterol (mmol/L)[d]				
Intervention group	1.07±0.32	1.07±0.25	1.20±0.29^{***y}	+0.12 (+0.06-+0.18)
Conventional treatment group	1.12±0.26	1.17±0.29	1.21±0.28	+0.05 (−0.01-+0.10)
Serum triglycerides (mmol/L)[e]				
Intervention group	2.76±1.60	2.50±1.44	1.96±0.89^{**y}	−0.5 (−0.91-−0.15)
Conventional treatment group	2.88±1.67	2.26±1.33^{**z}	2.33±1.19	+0.1 (−0.22-+0.37)

[a]Intervention group (n=40). Conventional treatment group (n=46).
[b]To convert mmol/L glucose to mg/dL, multiply mmol/L by 18.0. To convert mg/dL glucose to mmol/L, multiply mg/dL by 0.0555. Glucose of 6.0 mmol/L=108 mg/dL.
[c]To convert mmol/L cholesterol to mg/dL, multiply mmol/L by 38.9. To convert mg/dL cholesterol to mmol/L, multiply by 0.026. Cholesterol of 5.00 mmol/L=193 mg/dL.
[d]HDL cholesterol=high-density lipoprotein cholesterol.
[e]To convert mmol/L triglyceride to mg/dL, multiply mmol/L by 88.6. To convert mg/dL triglyceride to mmol/L, multiply mg/dL by 0.0113. Triglyceride of 1.80 mmol/L=159 mg/dL.
^{**z} $P<.01$, 0 vs 3 mo. ^{***z} $P<.001$, 0 vs 3 mo.
^{*y} $P<.05$, 3 vs 15 mo. ^{**y} $P<.01$, 3 vs 15 mo. ^{***y} $P<.001$, 3 vs 15 mo.
^{*x} $P<.05$, between the groups at 15 mo.

During the basic education period, serum total cholesterol and HDL cholesterol levels did not change, and serum triglycerides decreased significantly in the conventional treatment group. During the intervention period, serum triglycerides decreased ($P<.01$) and HDL cholesterol increased ($P<.001$) in the intervention group, but serum total cholesterol did not change in either group.

Diet

The mean energy intake tended to decrease in the intervention group and increase in the conventional treatment group (Table 3). Men in both groups had significantly higher intakes of energy, total fat, saturated, monounsaturated, and polyunsaturated fatty acids, and dietary cholesterol than women. Intakes of carbohydrate and fiber were significantly higher in women than in men.

At the end of the study, the main energy sources (when fiber was excluded from energy-containing nutrients) were cereals, meat products, and dietary fats, which were used as spreads and in cooking.

Intakes of total fat and monounsaturated fatty acids did not change in either group during the study. The intake of saturated fatty acids decreased in the intervention group and intake of polyunsaturated fatty acids increased in the conventional treatment group (Table 3). Fat, saturated fatty acids, and monounsaturated fatty acids were derived mainly from dietary fats, meat products, and milk products. The main sources of polyunsaturated fatty acids were dietary fats, meat products, cereals, and fish.

The mean dietary cholesterol intake remained within the recommended limits in both groups (Table 3).

Carbohydrates as a source of energy decreased significantly in the conventional treatment group, but there was no significant ($P=.077$) difference in the consumption of carbohydrates between the groups at the end of the study (Table 3). The main sources of carbohydrates were cereals, vegetables, milk products, and fruit. Only 2% to 3% of total energy came from sucrose. Fiber content of the diet did not change in either group during the study (Table 3). The main sources of fiber were cereals, vegetables, and fruit.

A significantly ($P<.05$) higher proportion of the intervention group achieved the recommendations for total fat (≤30% of total energy) and for saturated fatty acids (≤10% of total energy) (Figure 2). The proportion of patients in the conventional treatment group who had unsaturated fatty acid intakes of 20% of energy or more increased ($P<.05$) during the intervention period (Figure 2).

The recommendation for carbohydrate intake (>50% of energy) was achieved by 28% (10 of 38) of the patients in the intervention group and 22% (10 of 46) in the conventional treatment group. Only two patients in the intervention group and none in the other group consumed the recommended amount of dietary fiber (≥25 g/1,000 kcal) at the end of the study.

The intake of dietary cholesterol was lower than 300 mg in 66% (25 of 38) of the patients in the intervention group and in 72% (33 of 46) of those in the conventional treatment group at the end of the study.

Table 3
Intakes of energy, nutrients, fiber, and cholesterol at 3 and 15 months in the intervention and the conventional treatment groups

	Intervention group (n=38)		Conventional treatment group (n=46)	
	3 mo	15 mo	3 mo	15 mo
	mean ± standard deviation			
Energy[a] (kcal)				
Men	2,080±825	1,888±552	1,915±602	2,005±584
Women	1,331±585	1,283±459	1,130±308	1,214±350
Combined	1,710±773	1,628±580	1,633±636	1,713±630
Protein (% of energy)				
Men	19±2	19±3	19±4	18±3
Women	20±3	21±3	19±3	20±4
Combined	19±3	20±3	19±4	19±4
Fat, total (% of energy)				
Men	37±7	36±8	34±8	39±6**
Women	34±6	31±9	36±6	33±6
Combined	35±7	34±9	35±7	37±7
Saturated fatty acids (% of energy)				
Men	16±4	15±5	15±4	16±5
Women	13±6	12±5	15±3	13±2*
Combined	15±4	13±5*	15±4	15±4
Monounsaturated fatty acids (% of energy)				
Men	13±3	12±3	11±3	14±3**
Women	11±2	11±4	12±4	11±3
Combined	12±3	12±3	12±4	13±3
Polyunsaturated fatty acids (% of energy)				
Men	6±3	7±3	5±2	6±2*
Women	6±1	5±2	5±1	5±2
Combined	6±2	6±2	5±2	6±2*
Cholesterol (mg/day)				
Men	338±185	260±122	344±160	308±135
Women	208±116	215±110	194±72	198±71
Combined	273±166	237±117	285±151	265±126
Carbohydrates (% of energy)				
Men	41±7	42±5	44±9	39±9**
Women	45±7	48±8	44±5	45±8
Combined	43±7	45±7	44±7	41±9**
Sucrose (% of energy)				
Men	2±1	2±2	3±3	2±2
Women	3±2	3±1	3±1	3±2
Combined	2±2	3±2	3±3	2±2
Fiber (g/1,000 kcal)				
Men	16±5	16±4	14±5	14±4
Women	18±3	19±4	17±3	17±4
Combined	17±4	17±4	15±4	15±4

[a]Only digestible carbohydrates are included.
*$P<.05$, 3 to 15 mo. **$P<.01$, 3 to 15 mo.

Fatty Acid Composition of Serum Lipids

Palmitic acid of serum triglycerides decreased significantly ($P<.001$) during the study in the intervention group, and stearic acid increased during the basic education period (0 to 3 months) and decreased during the intervention period (3 to 15 months).

Similarly, palmitoleic acid of serum triglycerides decreased in the intervention group during the study but remained unchanged in the other group, while oleic acid decreased significantly ($P<.01$) in the conventional treatment group. Linoleic acid of serum triglycerides increased significantly ($P<.05$) in both groups during the intervention period (3 to 15 months).

The relative percentage of palmitic acid of serum triglycerides was lower and that of oleic acid was higher in the intervention group compared with the conventional group at the end of the study.

Palmitic acid of serum phospholipids decreased significantly in both groups during the study (Table 4). Monounsaturated fatty acids did not change in either group during the study. Linoleic acid of serum phospholipids increased significantly ($P<.05$) between the beginning and the end of the study in the intervention group and during the intervention period (3 to 15 months) in the other group.

The changes in monounsaturated fatty acids of serum cholesterol esters were a decrease in palmitoleic acid in the intervention group between the beginning and the end of the study and a decrease in oleic acid during the basic education period (0 to 3 months).

Of the polyunsaturated fatty acids, linoleic acid and α-linolenic acid of serum cholesterol esters increased in the intervention group between the beginning and the end of the study and in the conventional treatment group during the intervention period (3 to 15 months).

DISCUSSION

Loss of body weight and improved glycemic control before and during the basic education period suggest that the patients must have changed their dietary habits soon after the diagnosis. During the first two visits to the outpatient clinic of the university hospital, both groups of patients had the same basic dietary education, which was more extensive than usual in the clinical setting in Finland. This may explain the improvement in diabetic control during the basic education period.

All diabetic patients kept 4-day food records at three times during the study. In addition to giving an estimate of energy and nutrient intakes, the food records were used to help identify problem areas that required further dietary education of patients in the intervention group. Although feedback information from food records was given only to the intervention group, the recording task may have contributed to the weight loss in both groups. Recording food intake has been shown to influence the diet of the recorder and to predict weight loss (28). Food records made with household measurements are not as reliable for assessing intakes of nutrients as those made by weighing or by the duplicate portion technique (29).

Energy restriction and weight loss are the main principles involved in the treatment of obese patients with NIDDM (1-5). Even a slight weight loss may help improve glycemic control in patients with NIDDM (6,7). Weight loss in the intervention group from the time of diagnosis to the end of the study averaged about 7 kg. This result can be considered successful compared with earlier studies in which an average weight reduction of 2 to 5 kg was obtained (6,8,30-34). A moderate but continuous energy restriction during several months and changes in the dietary composition may have led to the weight loss, improved glycemic control, and improved serum lipid profile of patients in the intervention group. The patients in the conventional treatment group gained some weight, and metabolic control deteriorated to some extent during the intervention period.

The amount of carbohydrates and especially the amount of fiber in the diet were less than recommended. Only two of the patients had fiber intake of 25 g/1,000 kcal or more, which is recommended for obese patients with NIDDM (1). Cereals were the main source of fiber, so most of the fiber was nonsoluble. The recommended fiber content is difficult to achieve without special fiber products because legumes are not generally consumed in Finland.

Except for a study by Karlström and collaborators (11), studies have reported less promising results in regard to the composition of diet in patients with NIDDM, and the results have been far from recommendations (9,10,12,13,35,36). In the present study, the proportion of patients who fulfilled the dietary recommendations in respect to total fat and saturated fatty acids intakes was higher in the intervention group than in the conventional treatment group at the end of the study. Nevertheless, even in the intervention group only a small proportion of patients could achieve the strict goals. This finding is in accordance with many earlier studies (9-13,35,36) and suggests that the recommendations concerning diet for patients with NIDDM have been difficult to implement. It seems especially difficult to reduce the intakes of saturated fatty acids, total fat, and energy while increasing the intake of the unsaturated fatty acids. The main sources of saturated fatty acids and total fat in this study were dietary fats, milk, and meat products. To have fat intake in accord with recommendations, patients must choose low-fat meat and nonfat milk products and use moderate amounts of vegetable oils and margarines in cooking and as spreads.

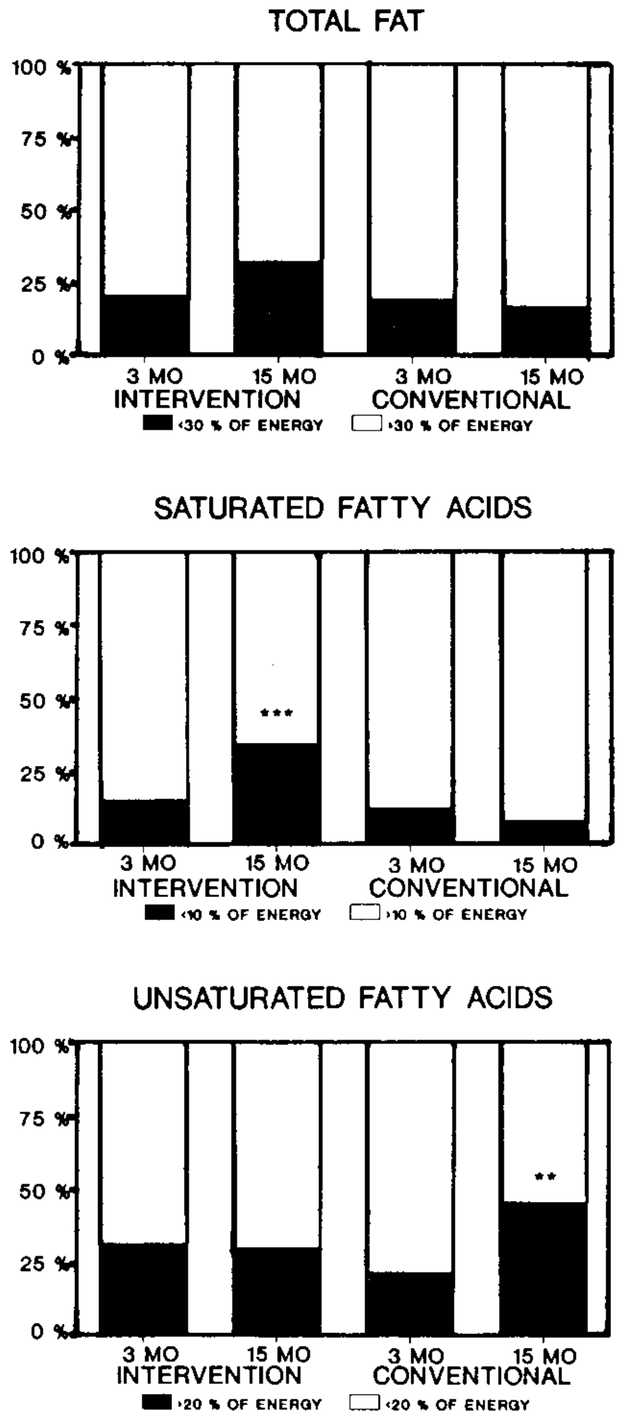

*FIG 2. Proportions of patients with diabetes who achieved the dietary goals for total fat, saturated fatty acids, and unsaturated fatty acids at 3 and 15 months of dietary therapy. There were significant differences between the groups in proportions of patients who achieved the goal for total fat (P<.05) and for saturated fatty acids (P<.001) at the end of the study. ***P<.001 and **P<.01 indicate within-group (3 vs 15 months) changes.*

Table 4
Fatty acid composition of serum triglycerides, phospholipids, and cholesterol esters in the intervention and conventional treatment groups at baseline, 3 months, and 15 months

Fatty acid	Intervention group			Conventional treatment group		
	0 mo	3 mo	15 mo	0 mo	3 mo	15 mo
	← mean ± standard deviation[a] →					
Serum triglycerides						
14:0 myristic acid	2.68±0.70	2.71±0.83	2.70±0.69	2.68±0.65	2.63±0.72	2.85±0.77
16:0 palmitic acid	31.85±2.63	31.43±2.65^{*z}	30.23±2.46^{***y}	31.60±2.85	30.81±3.26	31.55±3.55^{*c}
16:1 palmitoleic acid	5.77±1.39	5.67±1.29	5.49±1.41^{*y}	5.42±1.11	5.33±0.99	5.43±1.01
18:0 stearic acid	4.98±0.86	5.17±1.41^{*z}	4.75±0.76	5.12±0.99	5.01±0.94	4.89±0.72
18:1 oleic acid	39.81±2.31	39.96±3.31	40.36±3.20	40.35±2.91	40.45±3.11^{***z}	38.82±3.40^{**y*x}
18:2 *n*-6 linoleic	11.47±3.66	11.79±3.82^{*z}	13.32±3.47^{**y}	11.23±3.32	11.82±3.41^{*z}	12.94±4.75^{**y}
18:3 *n*-3 α-linolenic	0.82±0.42	0.68±0.52	0.80±0.69	0.83±0.46	0.71±0.52	0.81±0.66
Serum phospholipids						
14:0 myristic	0.60±0.15	0.58±0.14	0.56±0.14	0.57±0.11	0.59±0.12	0.57±0.16
16:0 palmitic	32.30±1.69	31.98±1.44^{*z}	31.56±1.50^{**y}	32.08±1.67	31.88±1.56^{*z}	31.47±1.93^{*y}
16:1 palmitoleic	1.37±0.44	1.33±0.41	1.29±0.43	1.25±0.30	1.27±0.28	1.31±0.38
18:0 stearic	14.74±1.41	14.71±1.17	15.07±1.55	14.88±1.46	14.87±1.20	14.98±1.12
18:1 oleic	14.38±2.17	14.31±2.23	14.18±2.07	14.33±1.86	14.30±1.75	14.36±2.28
18:2 *n*-6 linoleic	19.82±3.02	20.74±2.41	21.35±3.33^{*y}	19.45±2.44	19.24±3.17^{*z}	20.37±3.46
Serum cholesterol esters						
14:0 myristic	1.06±0.28	1.10±0.41	1.06±0.25	1.07±0.29	1.07±0.40	1.06±0.32
16:0 palmitic	13.00±1.06	13.03±0.97	12.82±1.24	12.95±0.91	13.21±1.23	13.16±1.06
16:1 palmitoleic	5.38±2.02	4.97±1.52	4.81±1.79	5.18±1.57	4.88±1.63	4.82±1.48
18:0 stearic	1.35±0.38	1.41±0.45	1.47±0.46	1.31±0.38	1.40±0.36	1.41±0.53
18:1 oleic	21.70±2.41	20.78±2.48	20.84±2.58	21.99±2.44	21.64±2.51	21.09±2.86
18:2 *n*-6 linoleic	46.19±5.54	47.58±4.98	48.37±5.44^{*y}	45.71±4.87	45.77±5.80	47.26±5.78
18:3 *n*-6 γ-linolenic	0.82±0.31	0.77±0.33^{*z}	0.89±0.40	0.88±0.42	0.79±0.39	0.86±0.35
18:3 *n*-3 α-linolenic	0.68±0.18	0.70±0.19^{**z}	0.80±0.23^{**y}	0.66±0.16	0.66±0.16^{*z}	0.73±0.19

[a]Values show relative percentages of each fatty acid in total fatty acids (mean ± standard deviation).
$^{*z}P<.05$, 3 to 15 mo. $^{**z}P<.01$, 3 to 15 mo. $^{***z}P<.001$, 3 to 15 mo.
$^{*y}P<.05$, 0 to 15 mo. $^{**y}P<.01$, 0 to 15 mo. $^{***y}P<.001$, 0 to 15 mo.
$^{*x}P<.05$, between the groups at 15 mo.

About two thirds (70%) of the patients in both groups achieved the goal for dietary cholesterol. Thus, the reduction of dietary cholesterol to recommended levels does not seem to be a major problem for outpatients with diabetes.

The fatty acid composition of serum triglycerides, phospholipids, and cholesterol esters reflects the quality of dietary fat (14-17). The increase in the percentage of linoleic acid of serum triglycerides, phospholipids, and cholesterol esters in the intervention group indicates that the dietary intake of linoleic acid was increased, because linoleic acid is derived from food only (37). The amount of linoleic acid in the diet has been shown to correlate well with its proportion in serum lipids (14-17). The decrease in the relative percentage of the palmitic fatty acid in all lipid fractions shows that intake of dietary palmitic acid decreased (38) in the intervention group. In the conventional treatment group, linoleic acid of serum triglycerides increased at the expense of oleic acid, and the increase of linoleic acid of phospholipids occurred at the expense of palmitic acid.

The same enzymes produce the desaturation and elongation of linoleic, α-linolenic, oleic, and palmitoleic acids (39). The competition between the substrates may partly explain the changes in the fatty acid composition of serum lipids. It has been suggested that the desaturation and elongation processes of fatty acids are impaired in insulin-dependent diabetes mellitus (39,40), but this has not been observed in patients with NIDDM. In the study carried out by Salomaa et al (18), the linoleic acid level of serum cholesterol esters was lower in patients with recently and previously diagnosed NIDDM than in nondiabetic subjects. On the other hand, the improvement in glycemic control was shown to increase the serum levels of the linoleic acid in one study of patients with NIDDM (41), but this finding is controversial (42).

Improved metabolic control might not be the sole explanation for the favorable changes in fatty acid composition of serum lipids in our study. The decreased intake of saturated fatty acids supports the fact that the quality of dietary fat of patients in the intervention group improved. The aforementioned changes in the composition of the dietary fat could contribute to a more favorable lipid profile (38) and to the better glycemic control observed in the intervention group at the end of the study. The increased use of n-6 polyunsaturated fatty acids and the decreased use of saturated fatty acids has been shown to improve insulin sensitivity in healthy people (43) and may also do so in patients with NIDDM.

APPLICATIONS

Even moderate energy restriction and slight changes in the amount and quality of dietary fatty acids, especially in favor of unsaturated fatty acids, can result in weight reduction and improvement in the glycemic control and the serum lipid profile of patients with recently diagnosed NIDDM. In most of these patients, good metabolic control can be achieved without drug therapy through intensive dietary intervention. ■

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